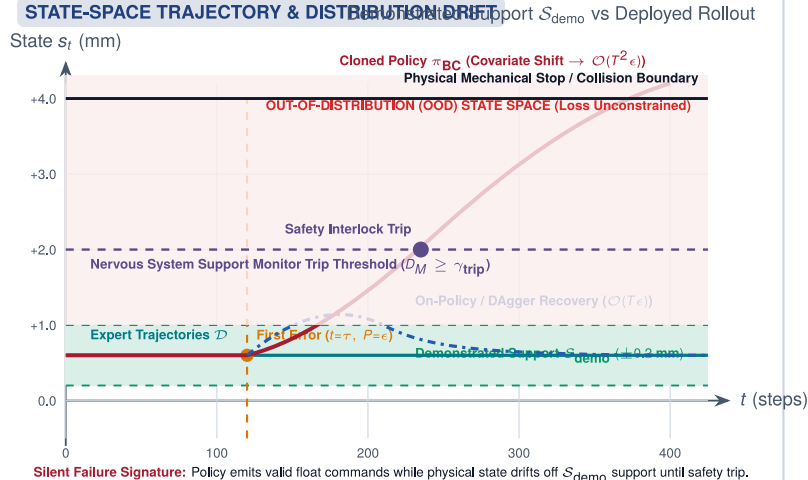


COMPOUNDING COVARIATE SHIFT AND STATE SUPPORT DRIFT IN BEHAVIORAL CLONING

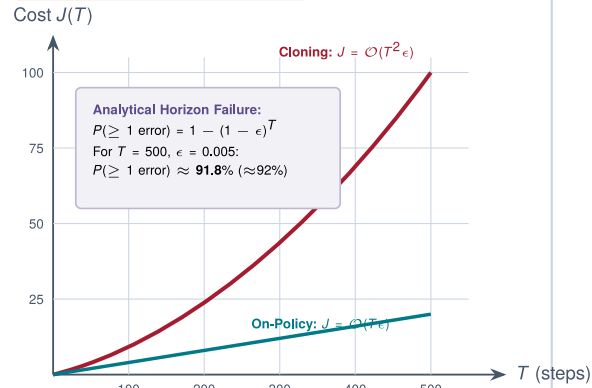
Closed-loop error propagation: single-step perturbation ϵ shifts physical state off demonstrated support, inducing $\mathcal{O}(T^2\epsilon)$ quadratic cumulative cost.

STATE-SPACE TRAJECTORY & DISTRIBUTION DRIFT



CUMULATIVE ERROR SCALING

Horizon T vs Cost J



Key Insight: Reducing ϵ postpones divergence time ($1/\epsilon$) but cannot remove $\mathcal{O}(T^2\epsilon)$ compounding.

CAUSAL FEEDBACK MECHANISM: FROM OFFLINE LOSS TO CLOSED-LOOP DESTRUCTION

Privilege Boundary & Endogenous Sensory Shift

1. LEARNED POLICY π_θ

- Trained on $\mathcal{D} = \{(o_t, a_t)\}$
- Zero recovery examples
- 1-step error probability ϵ
- Emits valid float commands

2. NERVOUS SYSTEM

- Privilege boundary check
- Commands look valid
- Monitor:** Mahalanobis D_M
- Passes command if unmonitored

3. PHYSICAL PLANT

- Actuator exerts force F_t
- Moves state: $s_t \rightarrow s_{t+1}$
- Endogenous Shift:**
 $s_{t+1} \notin \text{supp}(\mathcal{D}_{\text{demo}})$

4. SENSING

- Photons / Encoders
- Observation $o_{t+1} \sim p_\pi(o)$
- Covariate Shift:**
 $o_{t+1} \notin \text{supp}(\mathcal{D}_{\text{demo}})$

Closed-Loop Feedback: Unvisited Observation Query Off-Distribution $\Rightarrow$ Error Compounding $\mathcal{O}(T^2\epsilon)$